

Olga Petrovska, SFHEA

✉ olga.petrovska@swansea.ac.uk

🏠 Swansea, Wales, UK

🌐 opetrovska.github.io

🌐 olgapetrovska

🆔 0000-0003-1170-8816

Employment

- 2022 – 📌 **Head of Final Year** at Computer Science Department, Swansea University, UK
Interim Programme Director in 2022/23 (maternity cover)
- 2020 – 📌 **Teaching Fellow** at Computer Science Department, Swansea University, UK
- 2020 – 2025 📌 **Technology Consultant** and **Project Manager** at Ukraïner.net, remote (*part-time*)
- 2016 – 2020 📌 **Teaching Assistant** at Swansea University, UK (*part-time*)
- 2013 – 2015 📌 **Project Manager / Team Leader** at Wolfestone, Swansea, UK
- 2010 – 2012 📌 **Web Production Lead / Internal Relationships Leader** at IBM, Bratislava, Slovakia

Positions of Responsibility and Community Service

- 2025 – 📌 **Leader** of the “Embracing AI in Computer Science” Working Group within Swansea University’s CoSTA (Computer Science Teaching Attainment Group)
- 📌 **Co-leader** of the ITiCSE Working Group on Refactoring Upper Computing Courses to Leverage GenAI
- 📌 **Program Committee Member** at Human-Centered AI Education & Practice Conference (HCAI-ep’26), Computing Education Practice (CEP 2026), and Continuity, Computability, Constructivity (CCC 2025)
- 📌 **Reviewer** at Swansea University’s Advance HE-accredited Celebration of Professional Recognition (CoPR) programme
- 2024 – 📌 **Reviewer** for the several international journals (Frontiers in Education, Information and Software Technology, Media and Communication, and IEEE Access) and conferences (SIGCSE TS 2026, OpenEd25, Koli Calling’25, and BCSWomen Lovelace Colloquium)
- 2023 – 📌 **Council Member** at the Association of Computability in Europe
- 📌 **Project Support Officer** at the UK-Ukraine Academic Mentorship Scheme
- 2020 – 📌 **Treasurer** at the British Colloquium for Theoretical Computer Science

Education

- 2016 – 2021 📌 **Ph.D., Computer Science**, Swansea University, UK
Thesis: *Enhanced Realizability Interpretation for Program Extraction*
- 2015 – 2016 📌 **M.Sc., Computer Science**, Swansea University, UK
Thesis: *Prawf: Interactive Proof Assistant for Predicate Logic*
- 2006 – 2007 📌 **Master’s Degree, Philology**, Chernivtsi National University, Ukraine
Thesis: *The Functional-Semantic Field of Quantity in John Fowles’ Discourse*
- 2002 – 2006 📌 **Bachelor’s Degree, Pedagogy**, Ternopil National Pedagogical University, Ukraine

CPD Courses and Certifications

- 2025 📌 **Personal Development Course for External Examiners**, Advance HE

CPD Courses and Certifications (continued)

- Level 4 NVQ Diploma in Management, ILM
- Project Management Professional Certificate, Google
- Sustainability in the digital age: Energy-Efficient Software Development, openHPI
- Sustainability in the digital age: Environmental Impacts of AI Systems, openHPI
- 2024 ■ AI in Education: Leveraging ChatGPT for Teaching, University of Pennsylvania
- 2022 ■ Introduction to Quantum Computing with Qiskit (with IBM Quantum), openHPI
- Sustainable Software Engineering, openHPI

Teaching Experience

- 2025 – ■ Delivery of **CSB120MC Software Testing**
 - Design and delivery of **CSF1007 Professional Issues**
- 2021 – ■ Design and delivery of **CSF302 Project Planning and Management**
 - Design and delivery of **CSF206 Algorithms and Automata**
 - Coordination of final year project modules: **CSF301 Project Specification and Development** and **CSF300 Project Implementation and Dissertation**
 - Support of **CS-170/CS-175 Modelling Computing Systems**
- 2021 – 2025 ■ Designed and delivered **CSF107 Professional Issues of Software Engineering**
 - Designed and delivered a range of short courses and workshops for teachers, college students, and general public on **AI, Cyber Security, Graphic Design, Web Development, Excel**, and other.
- 2022 – 2023 ■ Delivered **CSB100MC Computational Thinking**
 - Delivered **CSB300MC Software Engineering Project Management**
 - Delivered components of **CS-175 Modelling Computing Systems 2**
 - Delivered components of **CSC301/CSCM01 Software Engineering Project Planning and Management**
- 2021– 2022 ■ Delivered components of **CSF399 Human-Computer Interaction**
- 2020 – 2021 ■ Delivered components of **CSF207 Computer Security**
 - Delivered **CSF106 Discrete Mathematics for Computer Science**
- 2016 – 2020 ■ Provided support with labs, tutorials and assessment in a variety of CS modules: **CS-205 Declarative Programming, CSCM13 Critical Systems, CSCM75 Logic in Computer Science, CS-275 Automata Theory and Formal Languages, CS-o81 Introduction to Algorithms and Data Structures.**

Publications

Conference Proceedings

- 1 O. Andrei, S. W. Nabi, M. Barr, and O. Petrovska, “Integrating socially responsible computing competencies in computer science and software engineering education,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, Association for Computing Machinery, Jan. 2025, pp. 36–37, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702225.

- 2 D. J. Bouvier, B. P. Cipriano, R. Glassey, R. Pettit, E. Anderson, A. Birillo, R. Dougherty, O. Hazzan, **O. Petrovska**, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, A. Vadaparty, and G. R. Yeluripati, “GenAI integration in upper-level computing courses,” in *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2*, ser. ITiCSE 2025, Nijmegen, Netherlands: Association for Computing Machinery, Jun. 2025, pp. 691–692, ISBN: 9798400715693.  DOI: 10.1145/3724389.3731276.
- 3 L. Clift and **O. Petrovska**, “Learning without limits: Analysing the usage of generative ai in a summative assessment,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, Association for Computing Machinery, Jan. 2025, pp. 5–8, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702214.
- 4 **O. Petrovska**, L. Clift, and F. Pantekis, “Video in assessments for soft skill development and evaluation,” in *Proceedings of the 2025 Conference on UK and Ireland Computing Education Research*, ser. UKICER’25, Association for Computing Machinery, Sep. 2025, ISBN: 9798400720789.  DOI: 10.1145/3754508.3754531.
- 5 **O. Petrovska**, R. Pearsall, and L. Clift, “Assessing software engineering students’ analytical skills in the era of generative ai,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, Association for Computing Machinery, Jan. 2025, p. 34, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702223.
- 6 **O. Petrovska**, L. Clift, F. Moller, and R. Pearsall, “Incorporating generative AI into software development education,” in *Proceedings of the 8th Conference on Computing Education Practice*, ser. CEP’24, Durham, United Kingdom: Association for Computing Machinery, Jan. 2024, pp. 37–40, ISBN: 9798400709326.  DOI: 10.1145/3633053.3633057.
- 7 **O. Petrovska**, L. Clift, and F. Moller, “Generative AI in software development education: Insights from a degree apprenticeship programme,” in *Proceedings of the 2023 Conference on United Kingdom & Ireland Computing Education Research*, ser. UKICER’23, Swansea, Wales UK: Association for Computing Machinery, Sep. 2023, ISBN: 9798400708763.  DOI: 10.1145/3610969.3611132.
- 8 U. Berger, **O. Petrovska**, and H. Tsuiki, “Prawf: An interactive proof system for program extraction,” in *Beyond the Horizon of Computability*, M. Anselmo, G. Della Vedova, F. Manea, and A. Pauly, Eds., Cham: Springer International Publishing, 2020, pp. 137–148, ISBN: 978-3-030-51466-2.  DOI: 10.1007/978-3-030-51466-2_12.
- 9 U. Berger and **O. Petrovska**, “Optimized program extraction for induction and coinduction,” in *Sailing Routes in the World of Computation*, F. Manea, R. G. Miller, and D. Nowotka, Eds., Cham: Springer International Publishing, 2018, pp. 70–80, ISBN: 978-3-319-94418-0.  DOI: 10.1007/978-3-319-94418-0_7.

Other

- 1 **O. Petrovska**, *Course Portfolio on Algorithms and Automata (Apprenticeship)*, 2024.  DOI: 10.5281/zenodo.14448266.
- 2 **O. Petrovska**, *Enhanced Realizability Interpretation for Program Extraction*, 2021.  DOI: 10.23889/suthesis.57831.

Selected Talks

- **Modular Generative AI Training to Promote AI Literacy Across the University** at the Swansea University annual Learning and Teaching Conference SUSALT 2025 - Enhancing Our Practice: Share - Learn - Create | co-presented with Filippos Pantekis (10.07.2025, Swansea University, Wales)
- **The Art of Teaching Theory Across Diverse Backgrounds** at the British Colloquium for Theoretical Computer Science 2025 (14-16.04.2025, University of Strathclyde, Scotland)

- **GenAI in Education and Research** at the NAIADES (Network for Artificial Intelligence, the Arts, the Digital Economy and Society) Workshop “GenAI: Use and Best Practices” (20.02.2025, Swansea University, Wales)
- **Computer Science Outreach in Higher Education: Challenges and Opportunities** | co-presented with Megan Venn-Wycherley (08.01.2025, Durham University, England)
- **Developing Students’ Critical Skills in the AI-enabled World** at the Advance HE Assessment and Feedback Symposium 2024 (05.11.2024, online)
- **How are students using Generative Artificial Intelligence?** (01.11.2024, “Pinch of SALT” podcast)
- **Generative AI and Conversations with Your Students** at the Lunch and Learn Webinar of the Swansea Academy of Learning and Teaching | co-presented with Lee Clift (22.10.2024, online)
- **Student vs AI** at the Learning and Teaching Enhancement Centre (LTEC) Conference “GenAI: Challenges and Solutions” (18.09.2024, Swansea University, Wales)
- **Operational Semantics in Prawf** at the Workshop on Proof and Computation 2022 (01.06.2022, Kloster Schlehdorf, Germany)
- **Intuitionistic Fixed Point Logic and Program Extraction** at the Workshop on Continuity, Computability, Constructivity 2018 (24-28.09.2018, University of Algarve, Faro, Portugal)
- **Natural Language Proofs for Program Extraction** at the International Workshop on Proof, Computation, Complexity 2017 (26-27.07.2017, University of Göttingen, Germany)
- **Natural Language Proof Checking in Predicate Logic** at the Workshop on Proofs, Programs, and Verification (19.04.2017, University of Canterbury, Christchurch, New Zealand)

Event Organisation

Sep 2025	■	CCC 2025 (Continuity, Computability, Constructivity) – Local organiser ¹
March 2025	■	UK-Ukraine Research Twinning Showcase Event 2025 at the Liverpool University – Co-organiser
Jul 2024	■	CCA 2024 (International Conference on Computability and Complexity in Analysis) – Local organiser
May 2024	■	Agda Implementors’ Meeting XXXVIII – Local organiser
Sep 2023	■	UKICER 2023 (UK and Ireland Computing Education Research) – Local organiser
Jul 2022	■	CiE 2022 (Computability in Europe: Revolutions and Revelations in Computability) – Local organiser
Apr 2022	■	BCTCS 38 (British Colloquium for Theoretical Computer Science) – Co-chair
Apr 2020	■	BCTCS 36 online conference – Local organiser
Sep 2019	■	2nd International Summer School on Proof Theory and Workshop on Proof Theory and its Applications – Local organiser

Secondments and Residences

Jul-Aug. 2018	■	Residence at the Hausdorff Trimester Program (Hausdorff Research Institute for Mathematics, Germany)
Mar-Apr. 2018	■	Secondment at the Mathematical and Information Sciences Division of the Human and Environmental Studies Faculty (Kyoto University, Japan)

¹All conferences/events organised at Swansea University, unless stated otherwise.

Secondments and Residences (continued)

Mar-May. 2017  **Secondment** at the School of Mathematics and Statistics (University of Canterbury, New Zealand)

Languages

Native or bilingual proficiency	 English, Ukrainian, Slovak, Russian
Limited working proficiency	 German, Polish
Elementary proficiency	 Welsh, Mandarin Chinese